UNIT 3 · TOPIC 11

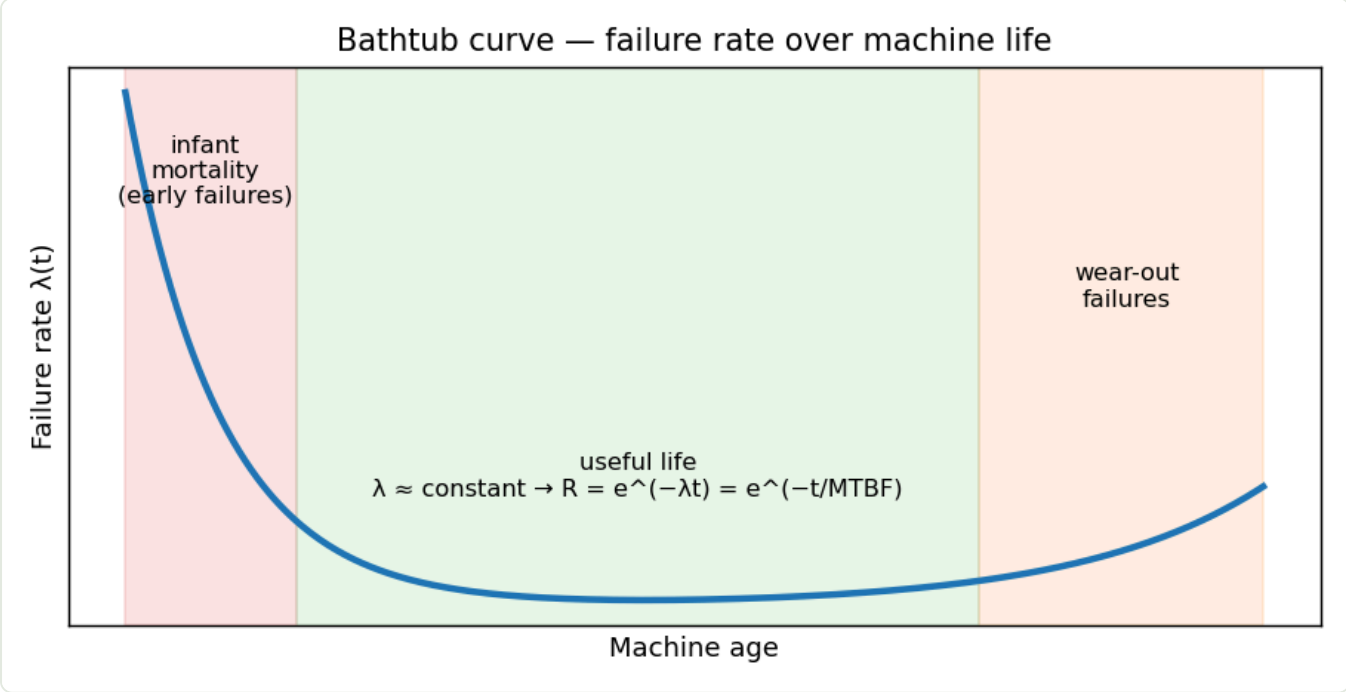
Reliability of Farm Machinery

Syllabus topic: Reliability of farm machinery

Reliability R(t) = the probability of performing the intended function for time t under stated conditions.

$$R(t) = e^{(-\lambda t)} = e^{(-t/MTBF)} \text{ (constant failure rate } \lambda \text{ — useful-life zone)}$$

- At t = MTBF, **R = 36.8%**.
- **MTBF** = mean time between failures (repairable machines); **MTTF** for non-repairable parts.



Bathtub curve — failure rate over machine life

Availability

$$\text{Availability} = MTBF/(MTBF + MTTR) \text{ (MTTR = mean time to repair)}$$

Maintainability matters as much as reliability in the season.

System reliability

System	Reliability
Series — most farm machines (any subsystem failure stops work)	$R_{sys} = R_1 \times R_2 \times \ldots$ — always LESS than the weakest component ($0.95^4 = 0.815$)
Parallel/redundant — rare (dual filters, twin knives)	$R_{sys} = 1 - \Pi(1 - R_i)$

Design lesson: fewer, robust components; easy field repair; carry critical spares in season.

> **Example.** MTBF 90 h, MTTR 10 h → availability = 90/100 = **90%**; over a 30-h operation, R = e^(-30/90) = **0.72**.

Practice MCQs

Q134. Reliability with constant failure rate is:

(a) R = 1-λt
(b) R = e^(-t/MTBF)
(c) R = t/MTBF
(d) R = λ²t

Q135. At an operating time equal to MTBF, reliability is about:

(a) 13.5%
(b) 36.8%
(c) 50%
(d) 63.2%

Q136. Four subsystems in series, each R = 0.95: system reliability ≈

(a) 0.95
(b) 0.86
(c) 0.815
(d) 0.75

Q137. Availability =

(a) MTTR/(MTBF+MTTR)
(b) MTBF/(MTBF+MTTR)
(c) MTBF×MTTR
(d) 1/MTBF

Q138. MTBF 90 h and MTTR 10 h give availability of:

(a) 80%
(b) 88%
(c) 90%
(d) 99%

Q139. The 'infant mortality' zone of the bathtub curve shows:

(a) Rising failures with wear
(b) High early failure rate that falls with run-in
(c) Constant failures
(d) No failures

Q140. Most farm machines behave as which reliability system?

(a) Parallel
(b) Series — any subsystem failure stops the machine
(c) Standby
(d) Redundant

Answers

Q134 — (b) R = e^(-t/MTBF)
Q135 — (b) 36.8% — *R = e^(-1) = 0.368.*
Q136 — (c) 0.815 — *$0.95^4 = 0.8145$.*
Q137 — (b) MTBF/(MTBF+MTTR)
Q138 — (c) 90% — *90/(90+10) = 0.90.*
Q139 — (b) High early failure rate that falls with run-in
Q140 — (b) Series — any subsystem failure stops the machine

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities and Cost Analysis); Hunt — Farm Power and Machinery Management; Jagdishwar Sahay; SMAM/CHC guidelines; NPTEL lectures.